

AMIR ZIASHAHABI

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Education

University of Southern California

Ph.D. in Electrical Engineering (GPA: 4/4)

Aug 2021 – Dec 2027 (Expected)

Los Angeles, CA

Sharif University of Technology

M.Sc. in Computer Engineering –Software (GPA: 19.38/20)

Sep 2017 – Jan 2020

Tehran, Iran

Iran University of Science and Technology

B.Sc. in Computer Engineering – Software (GPA: 17.66/20)

Sep 2013 – Sep 2017

Tehran, Iran

Research Interests

- Efficient Machine Learning
- Privacy-Preserving Machine Learning
- Large Language Models
- Distributed Systems

Research and Professional Experience

Machine Learning Researcher

vITAL Lab, University of Southern California (Advisor: Prof. Salman Avestimehr)

Aug 2021 – Present

Los Angeles, CA

- Conducting machine learning research with a focus on efficiency in system and algorithmic design.
- Areas of focus include large language models, federated learning, secure multiparty computation, distributed ML, and diffusion models.

Machine Learning Research Intern

TensorOpera

May 2026 – Aug 2026

Palo Alto, CA

- Developed post-training methods for large language models based on reinforcement learning and on-policy distillation

Machine Learning Research Intern

FedML, Inc.

May 2022 – Aug 2022

Los Angeles, CA

- Designed and developed hierarchical federated learning systems and analyzed communication backend performance.

Senior Developer & Distributed Systems Researcher

Rastak Media Sepehr Co.

Feb 2018 – Aug 2021

Tehran, Iran

- Led several projects, including developing a distributed ledger application with Hyperledger Fabric and a ride-hailing platform on Ethereum.
- Engineered solutions using Kubernetes, Solidity, React, NestJS and Web3 APIs.

Distributed Systems Researcher

Decentralab Lab, Sharif University of Technology (Advisor: Prof. Mohammad Ali Maddah-Ali)

Jan 2018 – Jan 2020

Tehran, Iran

- Investigated scalability solutions for blockchain with a focus on layer-2 solutions.
- Conducted a thesis on a blockchain-based social news aggregator.
- Contributed to curriculum design for a blockchain technology course.

Distributed Systems Researcher

Cloud Computing Center, Iran University of Science & Technology (Advisor: Prof. Mehrdad Ashtiani)

Dec 2016 – Sep 2017

Tehran, Iran

- Researched IaaS cloud computing solutions; deployed and monitored OpenStack services.

Technical Skills

Programming: Python, JavaScript, Solidity, CSS, HTML, Go, C#, PHP, Java, C/C++

Tools and Technologies: PyTorch RPC, gRPC, MPI, Docker, Kubernetes, Redis, Nginx, Unity Game Engine, Truffle

Libraries/Frameworks: PyTorch, Transformers, Diffusers, vLLM, GPT-Fast, Express, NestJS, React, Django, Flask, Laravel

Academic Honors

- Annenberg Fellowship, University of Southern California, 2021
- Ranked 2nd among M.Sc. students in Software Engineering, Sharif University, 2020
- Full-Tuition Scholarship, Ranked 14th among 25K+ in Nationwide M.Sc. University Entrance Exam, 2017
- Ranked 4th among B.Sc. students in Software Engineering, IUST, 2017
- Full-Tuition Scholarship, Top 1% of 251K+ in Nationwide B.Sc. University Entrance Exam, 2013

Selected Publications and Papers Under Review

Amir Ziashahabi*, Hossein Entezari Zarch*, Lei Gao, Murali Annavaram, Salman Avestimehr. ASPIRE: Asynchronous Batched Self-Speculative Decoding for Long-Context LLM Inference. *Conference on Language Modeling (COLM)*, 2026 (accepted).

Samuel Fernández-Menduiña*, **Amir Ziashahabi***, Eduardo Pavez, Antonio Ortega, Salman Avestimehr. Spend Bits Where Queries Look: KV Cache Vector Quantization with Attention-Preserving Transforms. *Under review*, 2026.

Narges Ghasemi, **Amir Ziashahabi**, Salman Avestimehr, Jonathan May. ORBIT: Training-Free Multi-Attribute Behavioral Steering via Orthogonal Subspace Rotation. *Under review*, 2026.

Amir Ziashahabi, Chaoyang He, Salman Avestimehr. Understanding the Communication Backends of Federated Learning, *IEEE International Conference on Communications (ICC)*, 2026.

Amir Ziashahabi*, Yavuz Faruk Bakman*, Duygu Nur Yaldiz, Mostafa El-Khamy, Sai Praneeth Karimireddy, Salman Avestimehr. Reject Only Critical Tokens: Pivot-Aware Speculative Decoding. *NeurIPS 2025 Efficient Reasoning Workshop*, 2025.

Lei Gao*, **Amir Ziashahabi***, Yue Niu, Salman Avestimehr, Murali Annavaram. MobiZO: Enabling Efficient LLM Fine-Tuning at the Edge via Inference Engines. *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

Narges Ghasemi*, **Amir Ziashahabi***, Salman Avestimehr, Cyrus Shahabi. GeoToken: Hierarchical Geolocalization of Images via Next Token Prediction. *IEEE International Conference on Data Mining (ICDM)*, 2025.

Amir Ziashahabi*, Narges Ghasemi*, Sajjad Shahabi, John Krumm, Salman Avestimehr, Cyrus Shahabi. OSMGen: Highly Controllable Satellite Image Synthesis using OpenStreetMap Data. *NeurIPS 2025 UrbanAI Workshop*, 2025.

Amir Ziashahabi, Baturalp Buyukates, Artan Sheshmani, Yi-Zhuang You, Salman Avestimehr. Frequency Domain Diffusion Model with Scale-Dependent Noise Schedule. *IEEE International Symposium on Information Theory (ISIT)*, 2024.

Hamza Saleem, **Amir Ziashahabi**, Muhammad Naveed, Salman Avestimehr. Hawk: Accurate and Fast Privacy-Preserving Machine Learning Using Secure Lookup Table Computation. *Privacy Enhancing Technologies Symposium (PETS)*, 2024.

Artan Sheshmani, Yi-Zhuang You, Baturalp Buyukates, **Amir Ziashahabi**, Salman Avestimehr. Renormalization Group flow, Optimal Transport and Diffusion-based Generative Model. *Physical Review E*, 2024.

Shen Li, Pritam Damania, Luca Wehrstedt, [et al., including **Amir Ziashahabi**]. PyTorch RPC: Distributed Deep Learning Built on Tensor-Optimized Remote Procedure Calls. *Proceedings of Machine Learning and Systems (MLSys)*, 2023.

*Equal contribution

Teaching Experience

Applied and Cloud Computing for Electrical Engineers — Teaching Assistant; University of Southern California

Foundations of Blockchain Technology — Teaching Assistant; Sharif University of Technology

Ethereum Programming Workshops — Instructor; Various Locations

Iran Blockchain Summer School — Co-organizer and Lecturer; Sharif University of Technology

Hands-on Blockchain Workshop — Co-organizer and Mentor; Sharif University of Technology

Database Design — Teaching Assistant; Sharif University of Technology

Compiler Design — Teaching Assistant; Iran University of Science and Technology

Selected Projects and Coursework

Pivot-Aware Speculative Decoding 🌀 (*Python, PyTorch, Transformers, vLLM, GPT-Fast*)

Speculative decoding that rejects only pivot-critical tokens to keep accuracy while boosting throughput.

GeoToken: Hierarchical Image Geolocalization 🌀 (*Python, PyTorch, Transformers*)

Geolocalization framed as next-token prediction for accurate location inference.

OSMGen: Controllable Satellite Image Synthesis 🌀 (*Python, PyTorch, Transformers, Diffusers*)

Satellite image generation conditioned on OpenStreetMap for structure-faithful, controllable outputs.

MobiZO: Efficient LLM Fine-Tuning at the Edge 🌀 (*Python, PyTorch, Transformers, PEFT, ExecuTorch*)

Edge-focused fine-tuning leveraging inference engines to remove backpropagation and cut VRAM on mobile/edge hardware.

FDDM: Frequency-Domain Diffusion 🌀 (*Python, PyTorch*)

Diffusion in the frequency domain with a scale-dependent noise schedule; achieving faster training and sampling.

Hierarchical Federated Learning 🌀 (*FedML, PyTorch, PyTorch RPC, MPI, gRPC, MQTT*)

Added hierarchical FL to FedML OSS stack by implementing distributed training primitives.

Software Fault Prediction (*R*)

Designed and evaluated a concise fault-prediction method on software-evolution data.

Toy Language + JVM Backend (*ANTLR, Jasmin*)

Built a small language with a compiler front-end and JVM bytecode backend.

Hunterguh: 2D Multiplayer (*Unity Game Engine, C#*)

Implemented 2D multiplayer action game.